

Juniorprogrammierer.de

Python 5: List

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List reminder I

```
1  #-----
2  #How to create a list
3  #-----
4  list1 = ["apple", "banana", "cherry"] # String
5  thislist = list(("apple", "banana", "cherry")) #different way
6
7  list2 = [1, 5, 7, 9, 3]
8  list3 = [True, False, False]
9  list4 = ["abc", 34, True, 40, "male"] # mixed
10
11 #-----
12 #How to access a list
13 #-----
14 print(list1[0])
15 print(list1[2])
16 #print(list1[3])
17 print(list4)
18 print(thislist)
19
20 #Print apple to orange
21 thislist = ["apple", "banana", "cherry", "orange", "kiwi", "melon", "mango"]
22 print(thislist[:4])
23
24 #Print starting cherry
25 thislist = ["apple", "banana", "cherry", "orange", "kiwi", "melon", "mango"]
26 print(thislist[2:])
27
28 #length of list
29 len(list1)
30
```

List reminder II

```
31 #-----
32 #How to manipulate a list
33 #-----
34 #add a value
35 list1.append("orange")
36
37 #remove a value
38 list2.remove(3)
39 list2.pop(1) # removes value of index 1
40 list2.pop() #remvoves last value
41
42 #Insert a value
43 list3.insert(1,True)
44
45 #Change 1 to n values of a list
46 thislist3 = ["apple", "banana", "cherry", "orange", "kiwi", "mango"]
47 thislist[1:3] = ["blackcurrant", "watermelon"]
48
49 #-----
50 #How to Loop throw a list
51 #-----
52 fruits = ["apple", "orange", "banana"]
53 for x in fruits:
54     print(x)
55
56 for i in range(len(fruits)):
57     print(x)
58
59 i = 0
60 while i < len(fruits):
61     print(fruits[i])
62     i += 1
63
```

List reminder III

```
80  #-----
81  #How to sort a list
82  #-----
83  numberList = [1,2,3,4,5]
84  numberList.sort() # sort numerical (aufsteigend)
85  print(numberList)
86  numberList.sort(reverse=True) #(absteigend)
87  print(numberList)
88
89  #-----
90  #How to create a copy of a list
91  #-----
92  oriList = [1,2,3,4]
93  copyList = oriList.copy()
```

Exercise normal array

Magical Creature Sanctuary Census

At the "Enchanted Creature Sanctuary," magical creatures are being cared for and monitored. The sanctuary recently conducted a census and recorded the number of magical sparkles each creature emits. Your task is to help the caretaker, Elora the Enchantress, analyze this data.

```
sparkle_counts = [23, 58, 31, 77, 85, 49, 92, 38, 67, 55]
```

- Help Elora to determine the following:
- What is the total number of sparkles emitted?
- What is the average number of sparkles per creature?
- What is the maximum sparkle count?
- What is the minimum sparkle count?
- List all creatures emitting more sparkles than the average sparkle count.
- List all creatures whose sparkle count falls within the range of 40 to 70 (inclusive), as they are part of the "Twilight Glow" group.